

SeatSense:

An AI-Powered Smart Predictor for College Admissions

Reducing uncertainty in college counselling through data-driven seat predictions, trend analysis, and personalised guidance.



The Problem Students Face



Counselling is broken for most students

Every year, thousands of students

navigate college admissions with little more than guesswork. Seat availability shifts daily, cutoff trends are opaque, and traditional predictors offer only surface-level comparisons.

→ Insufficient insights

No real-time visibility into seat availability or category-wise cutoffs

→ Overwhelming choices

Students shortlist poorly, wasting rounds on low-probability colleges

→ No personalisation

Generic tools ignore rank, category, and individual preferences



Introducing SeatSense

An AI-based system that predicts admission probability, analyses historical cutoff trends, and delivers personalised college suggestions — so students can make informed, confident choices.

Rank-Based Prediction

Admission probability scored
by entrance rank

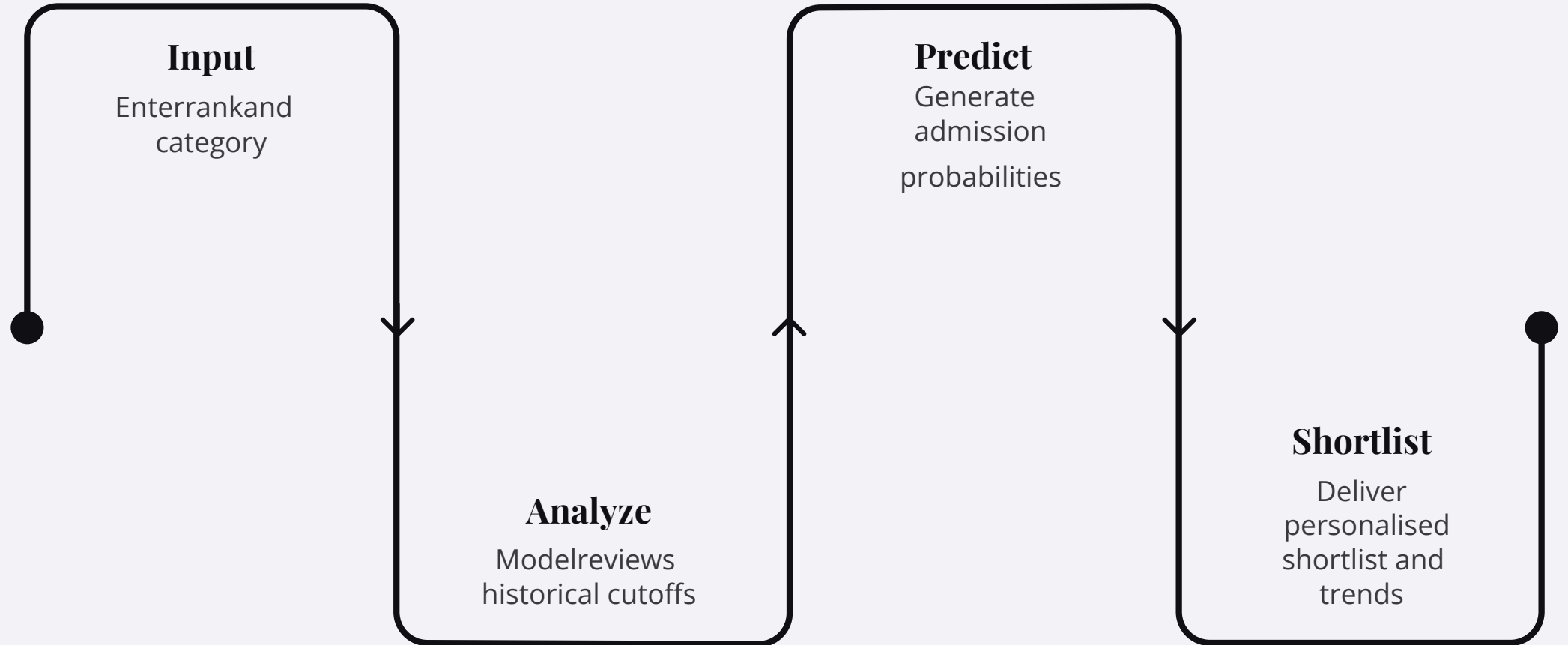
Trend Analysis

Historical cutoff patterns
surfaced clearly

Personalised

Category-aware, preference-matched suggestions

How SeatSense Works



SeatSense transforms raw entrance data into actionable intelligence — from input to personalised shortlist in seconds, powered by machine learning trained on multi-year admission records.

Core Features at a Glance



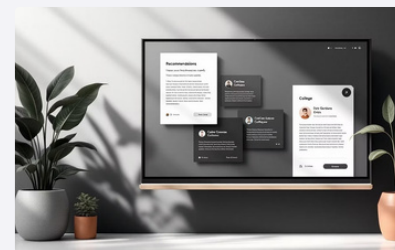
Cutoff Trend Analysis

Year-on-year cutoff visualisations help students understand how competitive each college truly is across rounds.



Admission Probability Score

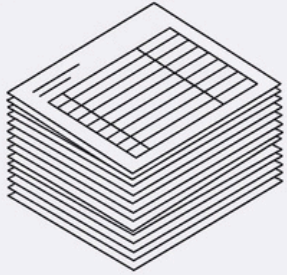
Each college receives a predicted probability score based on the student's rank, category, and seat availability patterns.



Smart Shortlisting

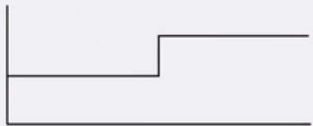
AI-curated college shortlists balancing ambition, safety, and realistic probability — tailored to every student's profile.

Why Traditional Predictors Fall Short



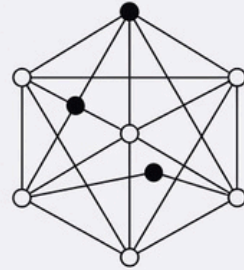
**STATIC CUTOFF
TABLES**

NO CATEGORY LOGIC



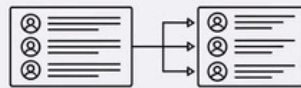
**NO PROBABILITY
SCORING**

**ONE-SIZE-FITS-ALL
OUTPUT**



**DYNAMIC ML
PREDICTIONS**

**CATEGORY-AWARE
LOGIC**



**PROBABILITY
SCORING**

**PERSONALISED
SHORTLISTS**

A fundamentally different approach

Most existing tools rely on static cutoff tables— a snapshot that becomes outdated the moment counselling rounds begin. SeatSense continuously analyses patterns, accounts for category reservations, and generates live probability estimates rather than binary yes/no comparisons.

- ❏ Prediction accuracy improves with each counselling round as more seat data becomes available.

The Technology Behind SeatSense

1

Data Collection & Preprocessing

Multi-year admission records, seat matrices, and category-wise cutoffs ingested and cleaned for model training.

2

Machine Learning Model

Regression and classification models trained to predict seat availability and admission probability per college-branch pair.

3

Personalisation Engine

Student profile (rank, category, preferences) mapped against predictions to generate ranked, contextual college suggestions.

4

Interactive Dashboard

Clean, accessible interface presenting probability scores, trend charts, and shortlists in one unified view.



Impact on Students & Counsellors



Better Decision-Making

Students apply strategically — fewer wasted rounds, higher chance of securing a preferred seat.



Reduced Anxiety

Data-backed predictions replace guesswork, giving students and families genuine confidence during counselling.

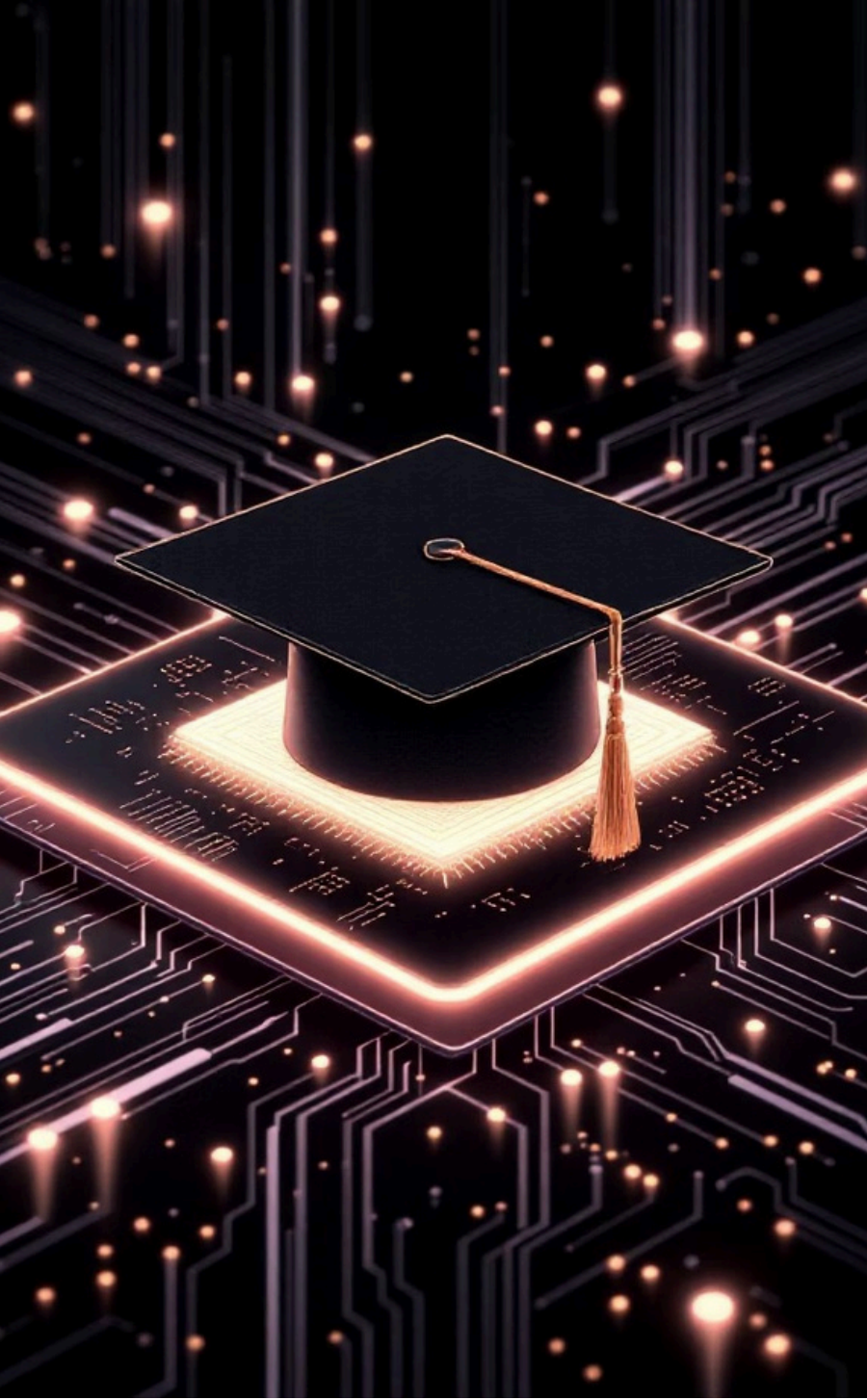


Empowered Counsellors

Counsellors gain a powerful analytical tool to guide students with evidence, not intuition alone.

Project Roadmap





The Future of College Counselling

SeatSense transforms one of the most stressful decisions in a student's life into a data-driven, guided experience. By combining AI with real admission intelligence, it empowers every student — regardless of background — to find the right college with clarity and confidence.

Predict Admission probability by rank & category	Analyse Multi-year cutoff trends at a glance
Decide Personalised shortlists backed by data	Chatbot Interact with our AI chatbot to resolve doubts, explore college options, and get smart recommendations tailored to your profile.